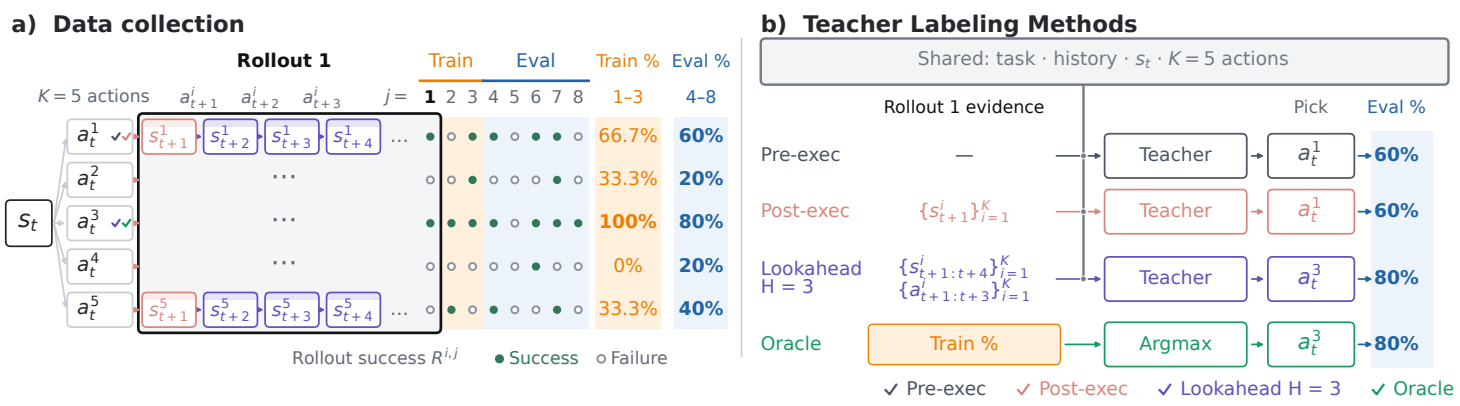
a) Data collection
Rollout 1
Train
Eval
Train %
Eval %
$K = 5$ actions
a^i_{t+1} a^i_{t+2} a^i_{t+3}
$j =$ 1 2 3 4 5 6 7 8
1–3
4–8
s_t
a^1_t s^1_{t+1} s^1_{t+2} s^1_{t+3} s^1_{t+4} ... 66.7% 60%
a^2_t ... 33.3% 20%
a^3_t ... 100% 80%
a^4_t ... 0% 20%
a^5_t s^5_{t+1} s^5_{t+2} s^5_{t+3} s^5_{t+4} ... 33.3% 40%
Rollout success $R^{i,j}$
Success
Failure
b) Teacher Labeling Methods
Shared: task · history · s_t · $K = 5$ actions
Rollout 1 evidence
Pick
Eval %
Pre-exec
—
Teacher
a^1_t
60%
Post-exec
$\{s^i_{t+1}\}^K_{i=1}$
Teacher
a^1_t
60%
Lookahead H = 3
$\{s^i_{t+1:t+4}\}^K_{i=1}$ $\{a^i_{t+1:t+3}\}^K_{i=1}$
Teacher
a^3_t
80%
Oracle
Train %
Argmax
a^3_t
80%
✓ Pre-exec
✓ Post-exec
✓ Lookahead H = 3
✓ Oracle